

Leveraging Mutual Intra-Modal Similarity Supervision for Text and Audio

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Abstract

Contrastive *inter*-modal representation learning is powerful in zero-shot classification, text-queried audio retrieval, and audio-queried text retrieval. Recently, image-text contrastive learning has been proposed with soft labels generated from text-text similarities between all samples in a batch. As our first contribution, we use such *intra*-modal text-text similarities as *inter*-modal text-audio similarity *targets*. Second, we generalize this to also gain information from audio-audio similarities. Third, we propose a method to obtain *similarity targets* from these *intra*-modal similarities using uncertainty estimation. We further investigate additionally employing a modality classification branch and using MSE-guided estimation. Finally, in a specified training regime, we show that our methods are competitive with SOTA on a number of downstream tasks, while using only a fraction of training data and batch size. Our utilized training data and pre-trained models are publicly available.

Index Terms: zero-shot classification, audio retrieval, text retrieval, representation learning, contrastive learning

1. Introduction

Predicting semantic similarities between different modalities is useful for many downstream tasks that deal with these modalities. The concept of inter-modal natural language supervision, first implemented in the contrastive language-image pre-training (CLIP) framework [1], heralded a breakthrough in this field. In CLIP, texts and images are transformed into a multi-modal embedding space, where the corresponding embeddings should be close together and far from all other embeddings, according to some distance metric. This was followed by an extension to joint image, language, and audio pre-training [2], and an adaptation for contrastive language-audio pre-training (CLAP) [3]. CLAP was improved by, e.g., different network topologies, more training data, and higher batch size, achieving SOTA performance on many downstream tasks (CLAP23) [4]. Some works enrich training by extracting additional information from metadata [5, 6], by augmenting texts [6–8], or by creating high-quality audio-text datasets [9, 10].

Other works improve the framework as a whole. A clustering loss is included in [11] to serve as an additional training signal. In [12], as a form of knowledge distillation, several trained models act as an ensemble in a second training stage to provide targets for a newly initialized model, instead of the normal one-hot targets. Sobal et al. [13] show that CLIP performance can be greatly improved by using estimated intra-modal text-text similarities for the texts in each batch as inter-modal targets.

In this paper, we propose to adopt text-text similarities as soft targets from Sobal et al. [13], and transfer this method from the image-text to the *audio-text field*. Second, we generalize the concept by also using intra-modal *audio-audio* similarities as inter-modal similarity *targets*. As a further contribution, we propose methods to combine the separate intra-modal similarities into joint inter-modal audio-text target similarities. As modality classifier branches with gradient reversal have improved performance in audio-text tasks [14, 15], we evaluate their benefit in conjunction with our proposals as a fourth contribution. As a further novelty, we propose to exchange the classification training paradigm by a regression paradigm, opting for a mean squared error (MSE) loss. *Lastly, in a specified training regime, we demonstrate that our methods are competitive with SOTA on three tasks: text-audio retrieval [16, 17], audio-text retrieval [17], and zero-shot audio classification [18, 19].*

2. Methods

2.1. Baseline System (purple area)

Figure 1 shows our overall training setup, including the baseline and our proposals. Here, we only focus on the baseline [12], of which we only use the first stage without knowledge distillation (purple color, solid arrows). Each batch is sampled as B corresponding audio-text pairs $(x_b(n), c_b)$, indexed by their batch-internal index $b \in \mathcal{B} = \{1, 2, \dots, B\}$, and n being the waveform sampling index. Each vector c_b represents a raw text. The audios and texts are brought to a shared, inter-modal embedding space through modality-specific encoders $\mathbf{E}^a$ and $\mathbf{E}^t$, including optional encoder-specific feature extractions and text tokenizers. The fully connected layers $\text{FC}(d)$ deliver d -dimensional embeddings. From the audio embeddings $\mathbf{Y} = (\mathbf{y}_b)^\top \in \mathbb{R}^{B \times d}$, with $\mathbf{y}_b \in \mathbb{R}^d$, and the text embeddings $\mathbf{Z} = (\mathbf{z}_b)^\top \in \mathbb{R}^{B \times d}$, with $\mathbf{z}_b \in \mathbb{R}^d$, an inter-modal audio-text similarity matrix $\mathbf{S} = (s_{i,j}) = (\text{sim}(\mathbf{y}_i, \mathbf{z}_j)) \in [-1, 1]^{B \times B}$ is calculated using the cosine similarity ($(\cdot)^\top$ is the transpose)

$$\text{sim}(\mathbf{a}_i, \mathbf{b}_j) = \frac{\mathbf{a}_i^\top \cdot \mathbf{b}_j}{\|\mathbf{a}_i\|_2 \cdot \|\mathbf{b}_j\|_2} \in [-1, 1], \quad (1)$$

with $\mathbf{a}_i, \mathbf{b}_j \in \mathbb{R}^d$ and $i, j \in \mathcal{B}$. Here, $\mathbf{a} = (\mathbf{a}_i)^\top = \mathbf{Y}$ and $\mathbf{b} = (\mathbf{b}_j)^\top = \mathbf{Z}$. If $\mathbf{a} = \mathbf{b}$, then $(\text{sim}(\mathbf{a}_i, \mathbf{b}_j))$ is symmetrical and its main diagonal all ones. $\mathbf{S}$ is not symmetrical, in general.

As further depicted in Figure 1, a symmetric target probability matrix $\bar{\mathbf{P}} \in \{0, 1\}^{B \times B}$ is defined with $\bar{P}_{i,i}=1$, $i \in \mathcal{B}$, and $\bar{P}_{i,j}=0 \forall i \neq j$ with $i, j \in \mathcal{B}$. Baseline [12] employs the loss

$$J^{\text{SP}}(\mathbf{S}) = \frac{1}{2} \left[J^\circ(\sigma(\mathbf{S}), \bar{\mathbf{P}}) + J^\circ(\sigma(\mathbf{S}^\top), \bar{\mathbf{P}}) \right], \quad (2)$$

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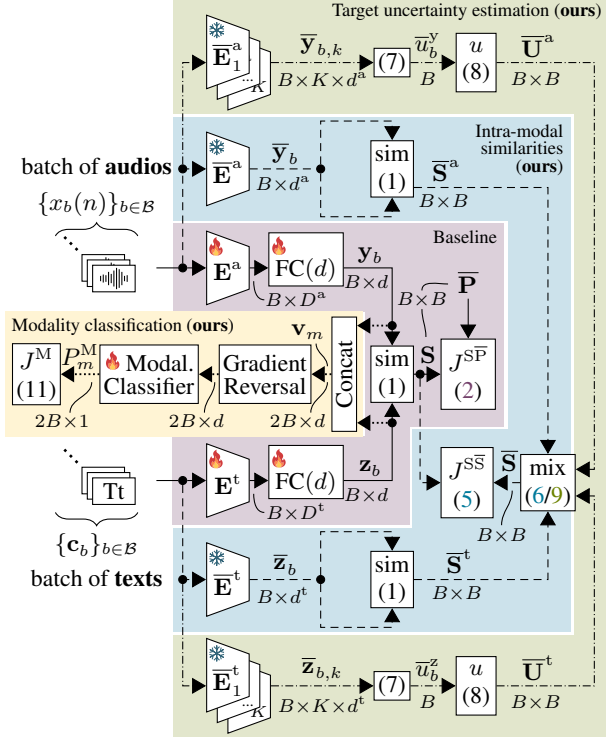


Figure 1: **Our training setup** with the baseline’s [12] first stage in the purple area. In the blue area, we produce intra-modal similarities and use them as soft targets. The green area’s uncertainty estimation pipelines enhance similarity mixing. The yellow area is optional, aiming to reduce the modality gap.

with $\mathbf{S} = (s_{i,j})$ and

$$\sigma_{i,j}(\mathbf{S}) = \frac{\exp(s_{i,j}/\tau)}{\sum_{k \in \mathcal{B}} \exp(s_{i,k}/\tau)} \in [0, 1], \quad (3)$$

defining the temperature-scaled row-wise softmax function $\sigma(\mathbf{S}) = (\sigma_{i,j}(\mathbf{S}))_{i,j \in \mathcal{B}}$, with $\mathbf{S} \in [-1, 1]^{B \times B}$. From (3), we obtain a probability matrix $\mathbf{P} = \sigma(\mathbf{S})$ with $\sum_{j \in \mathcal{B}} P_{i,j} = 1$ for all rows $i \in \mathcal{B}$. Note, $\sigma(\mathbf{S}^\top) \neq \sigma(\mathbf{S})$, in general. Temperature τ is a hyperparameter. The loss J° in (2) is set to the averaged (row-wise) cross-entropy (CE) loss

$$J^\circ(\mathbf{P}, \bar{\mathbf{P}}) = -\frac{1}{B} \sum_{i \in \mathcal{B}} \sum_{j \in \mathcal{B}} \bar{P}_{i,j} \log(P_{i,j}). \quad (4)$$

In (2), each row i (similarities of texts to one audio) and column j (similarities of audios to one text) of $\mathbf{S}$ is turned into a probability distribution and compared to the corresponding row or column of $\bar{\mathbf{P}}$, using the classification paradigm.

2.2. Proposed Intra-Modal Similarities (blue area)

Our first proposed method employs intra-modal similarities as soft targets for the inter-modal audio-text similarities. Such an approach has already yielded good results in the field of computer vision in predicting image-text similarities [13]. As shown in Figure 1 (blue), the intra-modal *text-text* similarity matrix $\bar{\mathbf{S}}^t \in [-1, 1]^{B \times B}$ is calculated by transforming the texts $\mathbf{c}_b$

into text embeddings $\bar{\mathbf{z}}_b \in \mathbb{R}^{d^t}$, $b \in \mathcal{B}$, via a frozen text encoder $\bar{\mathbf{E}}^t$, including its tokenizer, and calculating on a batch basis the cosine similarities (1) between them. We further generalize this approach by also producing *audio-audio* similarities $\bar{\mathbf{S}}^a \in [-1, 1]^{B \times B}$ with the use of a frozen audio encoder $\bar{\mathbf{E}}^a$, including its optional feature extraction. Both, $\bar{\mathbf{S}}^t$ and $\bar{\mathbf{S}}^a$, are symmetrical. To allow for similarity scores as targets, we define

$$J^{\text{SS}}(\mathbf{S}, \bar{\mathbf{S}}) = \frac{1}{2} \left[J^\circ(\sigma(\mathbf{S}), \sigma(\bar{\mathbf{S}})) + J^\circ(\sigma(\mathbf{S}^\top), \sigma(\bar{\mathbf{S}})) \right], \quad (5)$$

with $\sigma(\cdot)$ in analogy to (3), but with a different temperature $\bar{\tau}$, instead of τ . The so-called target similarity matrix $\bar{\mathbf{S}}$ is the result of a simple mixing operation (“mix” in Figure 1):

$$\bar{\mathbf{S}} = \lambda^a \cdot \bar{\mathbf{S}}^a + \lambda^t \cdot \bar{\mathbf{S}}^t, \quad (6)$$

which is a weighted average of the two intra-modal similarity matrices, with $\lambda^a \in [0, 1]$ and $\lambda^t = 1 - \lambda^a$ as weighting factors. Choosing (4) as J° , $\lambda^a = 0$, and $\lambda^t = 1$ yields the text-audio version of the text-image system proposed by Sobal et al. [13].

2.3. Proposed Target Uncertainty Estimation (green area)

As a further novelty, we present an enhanced mixing operation to obtain the target similarity matrix $\bar{\mathbf{S}}$ by estimating uncertainties of our intra-modal similarities, using input embedding uncertainties as proxies, which are computed using Monte Carlo dropout [20, 21] (Fig. 1, green area). As proposed in [22], we create K versions $\bar{\mathbf{E}}_k^a$, $k \in \mathcal{K} = \{1, 2, \dots, K\}$, of the encoder $\bar{\mathbf{E}}^a$, by randomly perturbing its parameters. For each input $x_b(n)$, $b \in \mathcal{B}$, the embedding uncertainty $\bar{u}_b^y \in [0, 1]$ is calculated as the sample circular variance [23, Section 2.3] of the perturbed encoders’ outputs $\bar{\mathbf{y}}_{b,k}$

$$\bar{u}_b^y = 1 - \frac{1}{K} \left\| \sum_{k \in \mathcal{K}} \frac{\bar{\mathbf{y}}_{b,k}}{\|\bar{\mathbf{y}}_{b,k}\|_2} \right\|_2 \in [0, 1]. \quad (7)$$

The more the *directions* of the embedding vectors $\bar{\mathbf{y}}_{b,k}$ spread out, the higher is the embedding uncertainty $\bar{u}_b^y$. Finally, as each target similarity $\bar{s}_{i,j}^a$, with $(\bar{s}_{i,j}^a) = \bar{\mathbf{S}}^a$, $i, j \in \mathcal{B}$ (cf. Section 2.2), is derived from embeddings $\bar{\mathbf{y}}_{b=i}$ and $\bar{\mathbf{y}}_{b=j}$, its uncertainty $\bar{u}_{i,j}^a$ is defined as the sum of their respective embedding uncertainties according to

$$\bar{u}_{i,j}^a = \bar{u}_{b=i}^y + \bar{u}_{b=j}^y, \quad (8)$$

where $(\bar{u}_{i,j}^a) = \bar{\mathbf{U}}^a$, $i, j \in \mathcal{B}$. In the lower text branch, target similarity uncertainties $\bar{u}_{i,j}^t$ are computed in analogy to (7), (8).

The enhanced target similarity matrix (formerly (6)) $\bar{\mathbf{S}} = (\bar{s}_{i,j})$ utilizes uncertainties as relative weights in its elements

$$\bar{s}_{i,j} = w(\bar{u}_{i,j}^t, \bar{u}_{i,j}^a) \cdot \bar{s}_{i,j}^a + w(\bar{u}_{i,j}^a, \bar{u}_{i,j}^t) \cdot \bar{s}_{i,j}^t, \quad (9)$$

and

$$w(a, b) = \begin{cases} \frac{a}{a+b}, & a+b > 0 \\ 0.5, & a+b = 0, \end{cases} \quad (10)$$

where $a, b \geq 0$ and always $w(a, b) + w(b, a) = 1$.

2.4. Proposed Modality Classification (yellow area)

From Sofer et al. [14], we utilize an optional branch with an auxiliary classifier and a gradient reversal layer (GRL) to shrink the modality gap (Fig. 1, yellow area). The GRL has no function

in the forward pass, but negates all gradients during backpropagation. The modality classifier individually processes each row $\mathbf{v}_m$ of its input $(\mathbf{v}_m)^\top = (\mathbf{y}_1, \dots, \mathbf{y}_B, \mathbf{z}_1, \dots, \mathbf{z}_B)^\top$ with $m \in \mathcal{M} = \{1, 2, \dots, 2B\}$, normalizing it to unit length and predicting an audio probability P_m^M , $m \in \mathcal{M}$, with $1 - P_m^M$ being the text probability. The averaged binary CE loss

$$J^M = -\frac{1}{2B} \sum_{m \in \mathcal{M}} [\bar{P}_m^M \log(P_m^M) + (1 - \bar{P}_m^M) \log(1 - P_m^M)] \quad (11)$$

is used, where $\bar{P}_m^M = 2 - \lceil \frac{m}{B} \rceil \in \{0, 1\}$ are targets.

2.5. Proposal: From CE to MSE

We noticed training instabilities when using small batch sizes where the system degraded to estimating all similarities as equal values. As part of a root cause analysis, we investigated loss landscapes and replaced the cross-entropy loss (4) used in (2) and (5) by the averaged mean squared error (MSE) loss

$$J^o(\mathbf{P}, \bar{\mathbf{P}}) = \frac{1}{B^2} \sum_{i \in \mathcal{B}} \sum_{j \in \mathcal{B}} (\bar{P}_{i,j} - P_{i,j})^2. \quad (12)$$

3. Experimental Setup

3.1. Datasets and Metrics

We evaluate model performance on tasks taken from [4], including zero-shot classification on ESC-50 [24], TUT Acoustic Scenes 2017 [25] (TUT17), NSynth [26], and Urban Sound 8K [27] (US8K), with accuracy (ACC) as the metric, as well as text-queried audio retrieval (TAR) and audio-queried text retrieval (ATR) on the ClothoV2.1 [28] evaluation split (Clotho-eval). For the latter two tasks the metrics are mean average precision on top 10 ranked assignments (mAP@10), and recall on the top ranked assignment (R@1), both reported using code from [9]. For TAR, we additionally report the mAP@16 with additional annotations (amAP@16) as provided by [29]. For each dataset metric, we assign a rank order to the methods. For each task, we report a rank accumulated across the datasets.

For training, we use the ClothoV2.1 [28] development split, AudioCapsV2 [30] dataset, and WavCaps dataset [9], and validate on the ClothoV2.1 [28] validation split (Clotho-val). We remove Clotho test data from WavCaps, according to [29].

3.2. Training

We build upon the framework by Primus et al. [12], available on github¹. The overall loss is defined as

$$J = J^S + \lambda^M \cdot J^M, \quad (13)$$

with $J^S = J^{S^P}$ (baseline, (2)) or $J^S = J^{S^S}$ (ours, (5)), J^M being the modality classification loss (11) and λ^M being a weighting factor for the modality classification loss (see Section 2.4).

According to Figure 1, we train $\mathbf{E}^a$, $\mathbf{E}^t$, the FC layers, and optionally the modality classifier for 20 epochs with an Adam optimizer. The learning rate schedule is a linear warmup from 0 to $2 \cdot 10^{-5}$ in the first epoch, followed by a half-cycle cosine ramp down to 10^{-7} over the following 13 epochs, whereafter the learning rate stays constant. The standard batch size is 64. We validate after each epoch and choose the checkpoint that delivers the highest mAP@10 score on ClothoV2.1-val. Following [13], the softmax temperature (3) is $\tau = 0.05$ and $\bar{\tau} = 0.1$.

¹<https://github.com/OptimusPrimus/salsa>

As audio encoder $\mathbf{E}^a$, we utilize PaSST [31] as in [12] without patch overlap, pre-trained on AudioSet [32] with the checkpoint version `passt_s-pl6-s16-128-ap468`, available on github². For training, we use structured patchout of two patches over frequency and 15 patches over time dimension. Each audio is segmented into 10 s snippets, or padded, if shorter. In case an audio is segmented, the resulting snippet embeddings are averaged to obtain one embedding for the entire file. As text encoder $\mathbf{E}^t$, we use pre-trained RoBERTa-large [33]. We have $d = 1,024$ and linear fully connected layers $\text{FC}(d)$.

For intra-modal similarity estimation (Section 2.2), audio encoder $\mathbf{E}^a$ is the same PaSST [31] model as $\mathbf{E}^a$, but kept frozen. Text encoder $\mathbf{E}^t$ is the pre-trained Sentence-BERT [34] variant `all-mpnet-base-v2` from HuggingFace³.

In uncertainty estimation (Sec. 2.3), we improve reproducibility and speed through replacing node-wise dropout by deterministic dropout layers, using $K = 10$. Each layer is created to output only zero values with a probability of P^{DO} , where P^{DO} is the original dropout layer’s dropout probability, or the output is the input multiplied with $\frac{1}{1 - P^{\text{DO}}}$. Original dropout probabilities of $P^{\text{DO}} = 0$ are changed to $P^{\text{DO}} = 0.1$. For models without dropout layers, we first append dropout layers with $P^{\text{DO}} = 0.1$ to each linear and each convolutional layer.

In modality classification (Section 2.4), the classifier is a simple five layer MLP. The first four halve their respective input dimensionality and use ReLU as activation function, whereas the last layer has one output node with sigmoid activation.

3.3. Evaluation

For datasets providing class labels, but no captions (all ZSC datasets), each instance’s text is created as “This is a sound of <class label>”, whereas otherwise the captions are used. For NSynth, we use instrument families as classes.

For all texts and all audios of a dataset, the respective embeddings are produced, and from them similarity matrix $\mathbf{S}$ (1). For inference, $\mathbf{S}$ is not necessarily quadratic. Each column describes similarities of one text to various audios, each row the reverse, with a higher value representing higher likelihood.

For zero-shot classification, the highest ranked text (and thereby class) for each audio is compared to ground truth. For text-queried audio retrieval (audio-queried text retrieval), the column (row) likelihoods are ranked and the order is evaluated.

4. Experimental Results and Discussion

Table 2 gives an overview of the trained models. Our loss (5) accepts similarities as targets, whereas (2) follows a one-hot approach. We compare CE (4) vs. MSE (12). Weights λ^a and λ^t determine the influence of the upper audio (lower text) branch similarities $\bar{\mathbf{S}}^a$ ($\bar{\mathbf{S}}^t$) on the final target similarities $\bar{\mathbf{S}}$, when using weighted mixing (6). For the enhanced mixing (9), the element-wise weighting is informed by the estimated uncertainties. By setting $\lambda^M = 0.01$, found by ablating on MSE+T+A+U+M, the modality classification branch “+M” is incorporated (13).

Table 1 shows performance on a zero-shot classification (ZSC) task in the first table segment, reporting accuracy (ACC) for all datasets. The second and third segments are text-queried audio retrieval (TAR) and audio-queried text retrieval (ATR). Here, the metrics introduced in Section 3.1 are employed. The first column shows the various SOTA methods and their

²<https://github.com/kkoutini/PaSST>

³<https://huggingface.co/sentence-transformers/all-mpnet-base-v2>

Table 1: Zero-shot classification (ZSC), text-queried audio retrieval (TAR), and audio-queried text retrieval (ATR) performance of each task’s SOTA (*values from papers), CLAP23 [4], our baseline [12] and our proposals. Per row, best is **bold**, second best underlined.

Batch size		various	1, 536	64													
#training samples		various	4.6M	0.5M (ours)													
Task	Test Dataset	Metric (%)	SOTA*	CLAP23 [4]	Baseline [12]	CE+T	CE+A	CE+T+A	CE+T+A+M	CE+T+A+U	CE+T+A+U+M	MSE+T	MSE+A	MSE+T+A	MSE+T+A+M	MSE+T+A+U	MSE+T+A+U+M
ZSC	ESC50	ACC	94.8 [9]	94.1	95.9	95.1	95.2	95.3	95.9	96.1	95.3	96.5	94.1	95.2	95.8	95.8	96.2
	TUT17	ACC	53.8 [4]	55.5	57.8	64.0	58.8	56.5	59.4	57.9	60.7	<u>62.0</u>	60.1	58.3	61.7	58.3	55.6
	US8K	ACC	82.3 [4]	82.6	80.1	82.3	81.3	81.8	<u>83.2</u>	82.3	82.5	83.4	82.6	82.4	80.4	80.0	81.1
	NSynth Instr.	ACC	47.9 [4]	44.7	33.1	34.7	34.0	32.8	33.0	34.7	34.5	32.4	35.4	34.5	33.2	35.7	<u>36.3</u>
	Accumulated rank ↓		—	26	34	<u>20</u>	29	35	22	23	<u>20</u>	16	21	26	26	28	25
TAR	Clotho-eval	mAP@10	41.7 [11]	25.8	32.3	33.6	33.2	32.6	32.5	32.8	33.7	34.4	34.4	34.4	<u>34.3</u>	<u>34.3</u>	34.2
	Clotho-eval	amAP@16	48.8 [11]	29.4	38.5	39.0	38.9	38.6	39.1	39.6	<u>40.0</u>	39.7	39.1	39.1	39.6	40.3	39.7
	Clotho-eval	R@1	28.4 [11]	16.0	21.3	22.4	21.8	21.2	21.2	21.3	22.1	<u>22.7</u>	22.8	<u>22.7</u>	<u>22.7</u>	<u>22.7</u>	<u>22.7</u>
	Accumulated rank ↓		—	29	25	14	18	23	21	19	10	6	7	8	6	5	8
ATR	Clotho-eval	mAP@10	15.5 [4]	15.6	19.8	19.2	19.2	18.5	19.1	19.4	18.3	<u>20.7</u>	20.1	20.8	<u>20.7</u>	19.5	20.0
	Clotho-eval	R@1	26.6 [14]	22.9	29.0	26.5	27.4	24.7	24.9	26.7	25.7	<u>29.1</u>	28.3	30.7	28.4	27.3	27.6
	Accumulated rank ↓		—	26	8	18	15	23	21	16	22	<u>4</u>	8	2	6	14	10

Table 2: Model variants used in this work, results are in Tab. 1.

Model name	J^S	J^o	λ^t	λ^a	λ^M	Mixing
Baseline [12]	(2)	(4)	—	—	0.0	—
CE+T	(5)	(4)	1.0	0.0	0.0	(6)
CE+A	(5)	(4)	0.0	1.0	0.0	(6)
CE+T+A	(5)	(4)	0.5	0.5	0.0	(6)
CE+T+A+M	(5)	(4)	0.5	0.5	0.01	(6)
MSE+T	(5)	(12)	1.0	0.0	0.0	(6)
MSE+A	(5)	(12)	0.0	1.0	0.0	(6)
MSE+T+A	(5)	(12)	0.5	0.5	0.0	(6)
MSE+T+A+M	(5)	(12)	0.5	0.5	0.01	(6)
CE+T+A+U	(5)	(4)	—	—	0.0	(9)
CE+T+A+U+M	(5)	(4)	—	—	0.01	(9)
MSE+T+A+U	(5)	(12)	—	—	0.0	(9)
MSE+T+A+U+M	(5)	(12)	—	—	0.01	(9)

reported performance employing various (partly proprietary) training datasets. As CLAP23 [4] is currently arguably the best *multi-task* zero-shot method with SOTA performance on a number of datasets, we separately inferred and report CLAP23 trained on 4.6 M training samples, using published weights. Although being first in ZSC on NSynth, overall, CLAP23 achieves only mediocre accumulated ranks of 26 ... 29 in a joint evaluation with our approaches.

Our methods are built upon baseline [12]. As the large batch size of 1,536 is hard to reproduce, we evaluate and compare our methods in a regime with batch size 64 and the same training data (0.5M training samples) as originally used in the baseline. Nevertheless, we also compare to CLAP23 with nine times more training data and 24 times the batch size.

From a bird’s eye view on accumulated ranks, we observe

that CE training (4) delivers only two second ranks, while our proposed MSE loss training (12) dominates with many strong results. Our best hypothesis is that in our case, the CE loss is struggling with soft targets. For the *ZSC task*, MSE+T yields a strong rank 16, confirming that intra-modal similarities serve as superior training targets compared to one-hot vectors. CE+T follows on rank 20 and MSE+A on rank 21, the latter revealing that also the audio modality yields good training targets. Further uncertainty estimation (+U) and/or modality classification (+M) does not help in ZSC. For the *TAR task*, MSE+T+A+U leads with a strong rank 5, confirming that uncertainty estimation makes sense in this task. Two further MSE methods closely follow on rank 6, which are MSE+T+A+M (including modality classification instead of uncertainty estimation) and MSE+T, again. Finally, for the *ATR task*, MSE+T+A is clearly the best choice with two top ranks, followed again by MSE+T.

Overall, MSE+T is the best multi-task method, as it is top in ZSC and second best in TAR and ATR. *Our MSE+T excels the baseline and CLAP23 in all tasks and all datasets except NSynth.* If optimal task performance is envisaged, task-specific settings can be chosen as discussed above.

5. Conclusions

In this paper, we leveraged mutual intra-modal similarity supervision for various downstream tasks. Our proposed methods achieve competitive performance to CLAP23, while using only a fraction of training data and batch size. We show that utilizing intra-modal text-text and audio-audio similarities for creating targets is beneficial in audio-queried text retrieval, while our additional uncertainty estimation excels in text-queried audio retrieval. Overall, our proposed MSE+T approach yields very strong multi-task performance and can be employed if task-specific optimization is not possible.

6. Use of Generative AI Disclosure

This manuscript was written without the use of Generative AI.

7. References

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